

CASE REPORT OPEN ACCESS

Efgartigimod as Add-On Treatment in Refractory Guillain–Barré Syndrome: A Case Report

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ABSTRACT

Background and Aims: Guillain–Barré Syndrome (GBS) is an autoimmune disease that can cause flaccid tetraplegia. Efgartigimod, a monoclonal antibody that inhibits the neonatal receptor (FcRn), increases IgG degradation.

Methods: A 60-year-old lady was admitted with acute flaccid tetraparesis, requiring mechanical ventilation within 12 h. Nerve conduction studies were compatible with acute inflammatory demyelinating polyneuropathy (AIDP). No anti-ganglioside or anti-nodal/paranodal antibody could be identified in serum. Despite receiving standard treatment with plasma exchange (PLEX), the patient remained severely tetraparetic, bedridden, and mechanically ventilated. Human immunoglobulin 2 g/kg was given, without obtaining significant improvement. One month after disease onset, the patient received efgartigimod (10 mg/kg intravenously) weekly for four doses.

Results: We report the results of this treatment on muscle strength, patient disability, and nerve conduction studies. The tolerability profile was satisfactory, and the patient was able to walk alone after 4 months from disease onset.

Interpretation: The potential role of efgartigimod in GBS should be explored in future clinical trials.

1 | Background and Aims

Despite the efficacy of standard treatments, that is, plasma exchange (PLEX) on alternate days for five exchanges or 2 g/kg intravenous immunoglobulin (IVIg), about 20% of Guillain–Barré syndrome (GBS) cases evolve to flaccid tetraparesis and need mechanical ventilation [1].

In such cases, available data do not support additional benefits from a second course of IVIg or PLEX, or from the combination of PLEX and IVIg.

Efgartigimod is a monoclonal antibody that binds the FcRn receptor and increases the degradation of IgG antibodies [2].

Efgartigimod has been approved for the treatment of IgG autoantibody-mediated diseases, such as myasthenia gravis and chronic inflammatory demyelinating polyneuropathy (CIDP).

2 | Case Report

A 60-year-old healthy woman, who received flu vaccination 7 days before, developed in 6 h a severe flaccid tetraparesis with generalized areflexia, stocking-glove hypoesthesia, progressive dysphonia, dysphagia, and respiratory failure. Within 12 h, the patient was bedridden and required mechanical ventilation in the Intensive Care Unit (Figure 1).

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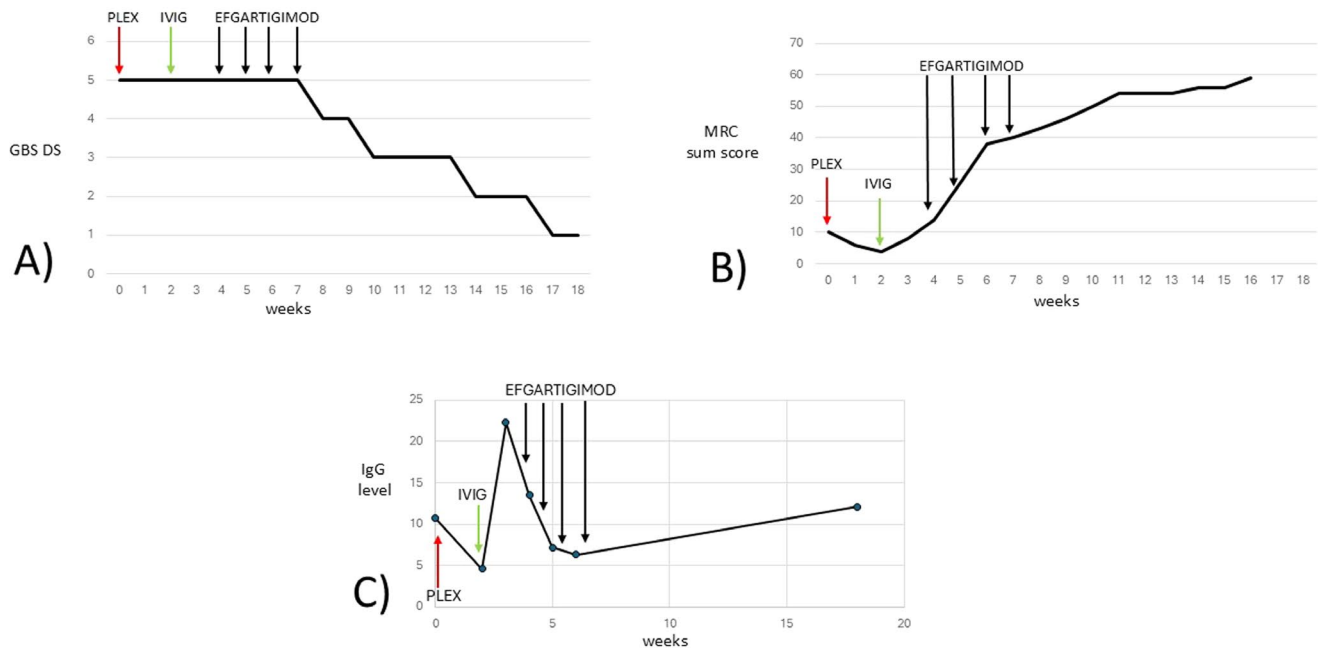


FIGURE 1 | Evolution of disease course, muscle strength, and serum IgG levels. The x-axis shows the time after admission (=week 0). The effect of different treatments on patient disability, represented by GBS disability score (GBS DS): 0=normal, 1=minor symptoms and capable of running; 2=able to walk 10 m or more without assistance but unable to run; 3=able to walk 10 m across an open space with help; 4=bedridden or chair bound; 5=requiring assisted ventilation for at least part of the day; 6=dead. The effect of different treatments on patient muscle strength, represented by the MRC sum score (range from 0 to 60). The nadir of weakness was reached after 2 weeks. Serum IgG level was normal (10.7 g/L) at admission, was reduced (4.6 g/L) after PLEX, increased after IVIg (22.3 g/L) and then progressively reduced after efgartigimod infusions (13.5 g/L, 7.2 g/L, 6.3 g/L). IgG level was normal again (12.2 g/L) after 10 weeks. PLEX=plasma exchange (i.e., 5 exchanges on alternate days). IVIG=intravenous immunoglobulin (0.4 g/kg/day for 5 days). MRC sum score: Range 0–60. IgG normal values: 7.37–16.07 g/L.

The patient's medical history was unremarkable, apart from sequestrectomy 2 months before for L5 radiculopathy.

Nerve conduction studies (NCS) (Figures S1 and S2) showed an increase in distal latencies, a reduction in conduction velocities, and reduced persistence of F waves. The sural nerve sensory action potential (SAP) was normal.

Cerebrospinal fluid analysis, performed 6 h after admission, showed normal findings, whereas whole spinal cord MRI showed gadolinium enhancement of the cauda equina.

The patient was admitted to the Intensive Care Unit because she was tetraplegic, requiring intubation, mechanical ventilation, and nasogastric tube for nutrition. The next morning, PLEX 3 L/session was started on alternate days, for five times (Figure 1). Tracheostomy was performed 8 days after admission, and gastrostomy was positioned 14 days after admission. At the end of PLEX treatment, the patient was still severely tetraparetic and mechanically ventilated (Figure 1).

Blood tests (WBC and RBC count, electrophoresis, reactive C protein) and serological tests (IgM and IgG for Cytomegalovirus, Epstein–Barr virus, *Mycoplasma Pneumoniae*, and Hepatitis E) were all negative. Serum collected at admission was tested for anti-ganglioside antibodies (GM-1 IgG and IgM, GT1a IgG and IgM, GD1a IgG and IgM, GD1b IgG and IgM, GQ1b IgG and IgM) and antibodies directed against nodal/paranodal structures (Caspr-1, Contactin-1, Neurofascin-155 and -186), with negative results.

At day 14, NCS showed inexcitable nerves in the lower limbs and severe reduction of compound muscle action potential (CMAP) of median and ulnar nerves (Figures S1 and S2). The SAP of median and ulnar nerves was bilaterally absent. Needle EMG showed spontaneous activity (fibrillations) in the right tibialis anterior muscle, with no voluntary activity in proximal muscles.

The next day (15 days after admission) we started IVIg 0.4 g/kg/day for 5 consecutive days. Nine days after the last IVIg infusion (1 month from admission), the patient was still severely tetraparetic, bedridden, and mechanically ventilated, with minimal improvement in the MRC sum score (Figure 1).

We thus started (at day 30 after disease onset) efgartigimod 10 mg/kg intravenously weekly for 4 cycles, observing already within a few hours an improvement in upper limb strength. The patient was transferred to the rehabilitation clinic.

After the second and the third infusion, strength in the upper limbs reached MRC score grade 4, but the patient was still bedridden. One week after the last efgartigimod infusion (2 months after admission), the tracheotomy could be removed, and 2 weeks later the patient walked with aid (Figure 1).

At 4 month follow-up visit, apart from absent deep tendon reflexes and mild stocking hypoesthesia, the neurological examination was normal, with normal strength in the four limbs (Figure 1). The same day, NCS showed marked improvement of CMAP amplitude of all nerves (Figures S1 and S2), whereas needle EMG did not detect any spontaneous activity.

Apart from a mild upper respiratory tract infection (successfully treated with co-amoxicillin for 5 days) after the second dose of efgartigimod, no other side effects were observed.

3 | Interpretation

Our patient suffered from a fulminant course of AIDP. Since the patient remained severely affected despite PLEX, immediately followed by IVIg, we discussed the off-label use of efgartigimod.

This treatment led to clinical improvement and was safe, but it is possible that we could have obtained the same clinical effect by repeating additional courses of PLEX (progressively reducing IgG levels) or waiting longer for the effect of the IVIg course.

Recently, in China, other severely affected GBS patients have been treated with efgartigimod as first-line treatment [3], or as sequential treatment after PLEX and IVIg [4–6], with a described favorable outcome. However, since the recovery trajectory of GBS is highly variable, it is impossible to draw any conclusion from such cases, and further data from studies with adequate design and sample size are warranted.

Author Contributions

P. Ripellino: conceptualization, investigation, funding acquisition, writing – original draft, methodology, writing – review and editing, data curation, supervision. **D. Dinacci:** conceptualization, investigation, data curation, supervision, writing – review and editing. **M. Conti:** conceptualization, investigation, data curation, writing – review and editing, supervision. **G. Galli:** writing – review and editing, data curation, methodology, investigation. **L. Frediani:** methodology, writing – review and editing, data curation, investigation. **P. Rossi:** methodology, visualization, data curation, investigation. **C. Gobbi:** investigation, writing – review and editing, methodology, supervision, writing – original draft.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1. Figure S1. Figure S2.**